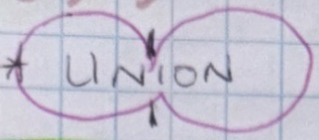
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EXERCISE 4
UNION AND UNION ALL

* UNION



Q1: Combine the account holders from both branches into one unique list. Each person should appear only once, even if they hold accounts at both branches.

Table: branch_sandton_accounts

account_id	account_holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi

Table: branch_rosebank_account

account_id	account_holder
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Motha

ANSWER:

```
SELECT account_id, account_holder, city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, city
FROM branch_rosebank_accounts;
```

account_id	account_holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Motha

Q2: Build a unique product catalogue that shows every product the bank offers from both the ~~products~~ current account and Savings account. Each product must appear once.

Table: Savings_Products

Product_code	Product_name
SV01	Basic savings
SV02	Premium savings
SV03	Youth Savings
SV04	Bussiness Savings

Table: current_products

Product_code	Product_name
CR01	Standard current
CR02	Gold current
CR03	Business Current.
SV02	Youth savings

ANSWER:

```
SELECT product_code, product_name FROM Savings_Products
UNION
SELECT product_code, product_name FROM current_products;
```

product_code	product_name	product_type
SV01	Basic Savings	Savings
SV02	Premium Savings	Savings
SV03	Youth Savings	Savings
SV04	Business Savings	Savings
CR01	Standard current	Current
CR02	Gold Current	Current
CR03	Business Current	Current.

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Q3: The HR team needs a unique list of all staff who work in either retail banking or corporate banking. Some staff are assigned to both. Each staff member should only appear once.

Table: retail_banking_staff

S001	Mpho Radebe
S002	Brian Tshabalala
S003	Aisha Patel
S004	Kabelo Moabelo

Table: corporate_banking_staff

S003	Aisha Patel
S005	Nandi Dube
S006	Sipho Khumalo
S004	Kabelo Moabelo

Answer:

```
SELECT staff_id, staff_name, email FROM retail_banking_staff
UNION
SELECT staff_id, staff_name, email FROM corporate_banking_staff;
```

staff_id	staff_name	email
S001	Mpho Radebe	mpho.radebe@nexbank.co.za
S002	Brian Tshabalala	brian.t@nexbank.co.za
S003	Aisha Patel	aisha.p@nexbank.co.za
S004	Kabelo Moabelo	kabelo.m@nexbank.co.za
S005	Nandi Dube	nandi.d@nexbank.co.za
S006	Sipho Khumalo	sipho.k@nexbank.co.za.

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Q4: Marketing needs a unique list of all cities where NexBank serves customers - either through its mobile branch or through its digital branch (which serves customers online regardless of city). No city should appear twice.

Table: mobile_branch_cities

city_code	city_name
C01	Johannesburg
C02	Pretoria
C03	Cape Town
C04	Durban

Table: digital_branch_cities

city_code	city_name
C03	Cape Town
C05	Polokwane
C06	Port Elizabeth
C01	Johannesburg

ANSWER:

```
SELECT city_code, city_name, region FROM mobile_branch_cities
UNION
SELECT city_code, city_name, region FROM digital_branch_cities;
```

city_code	city_name	Region
C01	Johannesburg	Gauteng
C02	Pretoria	Gauteng
C03	Cape Town	Western Cape
C04	Durban	KwaZulu Natal
C05	Polokwane	Limpopo
C06	Port Elizabeth	Eastern Cape

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Q5: The digital marketing team ran two campaigns last month - one via push notifications, and one via in-app banner. Build a unique list of every customer who was targeted. If a customer received both, they must appear only once.

Table:
Push_notifications_targets

customer_id	customer_name
C1001	Nomsa Zwane
C1002	Andile Buthelezi
C1003	Fatima Mohamed
C1004	Ryno Van Zyl

Table:
inapp_banner_targets

customer_id	customer_name
C1003	Fatima Mahamed
C1005	Thandeka Cele
C1006	Samuel Nkosi
C1002	Andile Buthelezi

ANSWER:

```
SELECT customer_id, Customer_name, segment From push_notification_targets
UNION
SELECT customer_id, customer_name, Segment From inapp_banner_targets;
```

customer_id	customer_name	Segment
C1001	Nomsa Zwane	Retail
C1002	Andile Buthelezi	Retail
C1003	Fatima Mohamed	SME
C1004	Ryno Van Zyl	Corporate
C1005	Thandeka Cele	Retail
C1006	Samuel Nkosi	Private Wealth.

* QUESTION 6 - 10 (UNION ALL)

Q6: The fraud team needs the complete log of all ATM transactions from machine ATM_01 and machine ATM_02. Every transaction must appear - even if the same transaction appears in both logs due to a system sync error.

Table: atm_01_transactions

txn_id	account_id
T1001	A001
T1002	A002
T1003	A001
T1004	A003

Table: atm_02_transactions

txn_id	account_id
T1003	A001
T1005	A004
T1006	A002
T1007	A005

ANSWER:

```
SELECT transaction_id, account_id, amount, transaction_date
FROM atm_01_transactions
UNION ALL
SELECT transaction_id, account_id, amount, transaction_date
FROM atm_02_transactions;
```

transaction_id	account_id	amount	transaction_date
T1001	A001	500.00	2026-03-01
T1002	A002	120.00	2026-03-01
T1003	A001	300.00	2026-03-02
T1004	A003	750.00	2026-03-02
T1003	A001	300.00	2026-03-02
T1005	A004	900.00	2026-03-03
T1006	A002	450.00	2026-03-03
T1007	A005	150.00	2026-03-04

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Q7: The credit team wants a full list of all loan applications from the Gauteng region and Western Cape region. Every application must be included - duplicates are allowed because both regions may have captured the same application.

Table: gauteng_loan_applications

app_id	customer_id
LA001	C1001
LA002	C1002
LA003	C1003
LA004	C1004

Table: western_cape_loan_applications

app_id	customer_id
LA003	C1003
LA005	C1005
LA006	C1006
LA007	C1007

ANSWER:

```
SELECT application_id, customer_id, loan_type, amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT application_id, customer_id, loan_type, amount_requested
FROM western_cape_loan_applications;
```

application_id	customer_id	loan_type	amount_requested
LA001	C1001	Home_loan	850000,00
LA002	C1002	Vehicle loan	220000,00
LA003	C1003	Personal	45000,00
LA004	C1004	Home Loan	1150000,00
LA003	C1003	Personal	45000,00
LA005	C1005	Vehicle Loan	310000,00
LA006	C1006	Business	550000,00
LA007	C1007	Home Loan	920000,00

Q8: The customer experience team needs a full record of every complaint logged this month, wether it came via email, or via the mobile app. Every complaint must appear - The team needs all of them to calculate total complaints volume.

Table: email_complaints

complaint_id	customer_id
EC001	C2001
EC002	C2002
EC003	C2003
EC004	C2004

Table: app_complaints

Complaint_Id	customer_Id
AC001	C2005
AC002	C2001
AC003	C2006
AC004	C2007

ANSWER:

```
SELECT complaint_id, customer_id, category, logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id, customer_id, category, logged_date
FROM app_complaints;
```

complaint_id	customer_id	category	logged_date
EC001	C2001	Online Banking	2026-03-05
EC002	C2002	ATM failure	2026-03-06
EC003	C2003	Card dispute	2026-03-07
EC004	C2004	Account Fees	2026-03-08
EC0081	C2005	App crash	2026-03-05
EC002	C2001	Card dispute	2026-03-06
EC003	C2006	Payment delay	2026-03-07
EC004	C2007	Login issue	2026-03-08

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Q9: Finance needs to reconcile all payment records from April and May into one table for the half-year report. Every payment from both months must appear — the total row count matters for reconciliation.

Table: April_Payments

payment_id	account_id	amount
PAY001	A001	12500,00
PAY002	A002	4800,00
PAY003	A003	9200,00
PAY004	A004	3300,00

Table: May_Payments

Payment Id	account_id	amount
PAY005	A001	12500,00
PAY006	A005	7600,00
PAY007	A002	5100,00
PAY008	A006	2800,00

ANSWER:

```
SELECT payment_id, account_id, amount FROM april_payments
UNION ALL
SELECT payment_id, account_id, amount FROM May_payments;
```

payment_id	account_id	amount	payment_date
PAY001	A001	12500,00	2026-04-15
PAY002	A002	4800,00	2026-04-18
PAY003	A003	9200,00	2026-04-25
PAY004	A004	3300,00	2026-04-28
PAY005	A001	12500,00	2026-05-15
PAY006	A005	7600,00	2026-05-20
PAY007	A002	5100,00	2026-05-22
PAY008	A006	2800,00	2026-05-27

Q10: The accounting system records debits and credits in two separate tables. Combine all entries into one general ledger for the audit. Every single entry from both tables must appear - no rows should be dropped.

Table: debit_entries

entry_id	account_id	entry_type
DR001	A001	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit

Table: credit_entries

entry_id	account_id	entry_type
CR001	A001	Credit
CR002	A005	credit
CR003	A002	credit
CR004	A006	credit.

ANSWER:

```
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM credit_entries;
```

entry_id	account_type	entry_type	amount	entry_date
DR001	A001	Debit	1500,00	2026-03-10
DR002	A002	Debit	3400,00	2026-03-11
DR003	A003	Debit	850,00	2026-03-12
DR004	A004	Debit	2100,00	2026-03-13
CR001	A001	Credit	5000,00	2026-03-10
CR002	A005	credit	1200,00	2026-03-11
CR003	A002	Credit	4500,00	2026-03-12
CR004	A006	Credit	3100,00	2026-03-13

BONUS QUESTIONS

Q1: A Fraud analyst wants a unique list of customers who appeared in either the January or February high risk watch lists. Should they use UNION or UNION ALL? Explain

ANSWER:
They must use UNION because the question clearly states that ① It is a unique list, ② ~~they~~ the Customer must apper in either The Jan or Feb. They do not want customers who appeared in both. The UNION function fillers duplicates.

Q2: The auditing team wants to count every individual transaction that passed through the bank in Q1 - even if the same transaction was recorded in two systems on the same day. Should they use UNION or UNION ALL? why?

ANSWER
They must use UNION ALL because this function includes all rows including duplicates from all queried systems. This function allows the user to satisfy the requirement that all transactions must be included even if they are the same in two different systems.

Q3: A developer writes this query: SELECT account_id, account_holder FROM branch_condton_accounts UNION SELECT account_id, account_holder, city FROM branch_rosebank_accounts. The query fails. What is wrong and how would you fix it?

ANSWER: The number of columns selected from the first table does not match those selected from the second table. To fix: include city in the Select statement from the first statement, or exclude city from the second select statement